TOPIC 01

Regression Analysis and Correlation vs Causation

What regression is for, how correlation measures a relationship, and why a strong correlation does not prove one thing causes another

WHAT YOU WILL LEARN

- Explain what regression analysis is used for
- Name the dependent and independent variable in a situation
- Recall and compute the correlation coefficient, and read its value
- Use the Vin Diesel example to see a coincidental correlation
- Name three reasons a correlation can appear without causation
- Explain why a small sample makes correlations unreliable

NOTE **Where this connects**

The inferential course ended with hypothesis tests and p-values (Topics 10, 12 and 13). Regression is where those ideas get used on real relationships between variables. It also builds on the **correlation coefficient** from descriptive statistics, which this topic recalls first.

1. What regression analysis is for

Regression analysis is the most common method for **prediction** and for describing how one variable depends on others. It is used whenever there is a plausible cause-and-effect relationship between variables.

Outcome we want to understand (dependent variable y)	What may drive it (independent variable x)
Money you spend	Money you make
Time you devote to studying statistics	Your motivation
College GPA	SAT score
House price	Size, location, year built

- The **dependent variable (y)** is the thing being explained or predicted. It is also called the response or outcome.
- The **independent variable (x)** is the thing used to explain or predict it. It is also called the predictor, regressor or explanatory variable.
- The arrow of influence runs from x to y: your income affects your spending, not the other way round.

Roadmap of this part of the course

1. The **simple linear regression** model (one x).
2. How to **build** a regression: the line and least squares.
3. How to **interpret** a regression: R^2, tables, coefficients.

4. How to **compare** models (adjusted R^2).
5. The **multivariate** linear regression model (many x).
6. After this: advanced statistical analysis and machine learning, where regression is a basic building block.

2. Correlation: a quick recap

Before regression we need a way to tell whether two variables move together. A **scatter plot** shows the pattern; the **correlation coefficient** puts a number on it.

$$r = \frac{s_{xy}}{s_x s_y} \quad \text{(sample)} \qquad \rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \quad \text{(population)}$$

Here s_{xy} is the sample **covariance** of x and y, and s_x, s_y are the standard deviations. Dividing by both standard deviations removes the units, so r always lies between −1 and +1.

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Covariance is the average of the products of the two deviations from the means. If x and y tend to be above their means together (or below together), the products are mostly positive and s_{xy} is positive; if one is high when the other is low, it is negative. Covariance has awkward units (x-units times y-units), which is why we divide to get r.

A five-point example by hand. Take x = 1, 2, 3, 4, 5 and y = 2, 4, 5, 4, 5, so $\bar{x} = 3$ and $\bar{y} = 4$.

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
1	2	−2	−2	4	4	4
2	4	−1	0	−0	1	0
3	5	0	1	0	0	1
4	4	1	0	0	1	0
5	5	2	1	2	4	1
Sum				6	10	6

$s_{xy} = 6/4 = 1.5$; $s_x = \sqrt{(10/4)} = 1.581$; $s_y = \sqrt{(6/4)} = 1.225$. So $r = 1.5/(1.581 \times 1.225) =$ **0.775**: a fairly strong positive correlation.

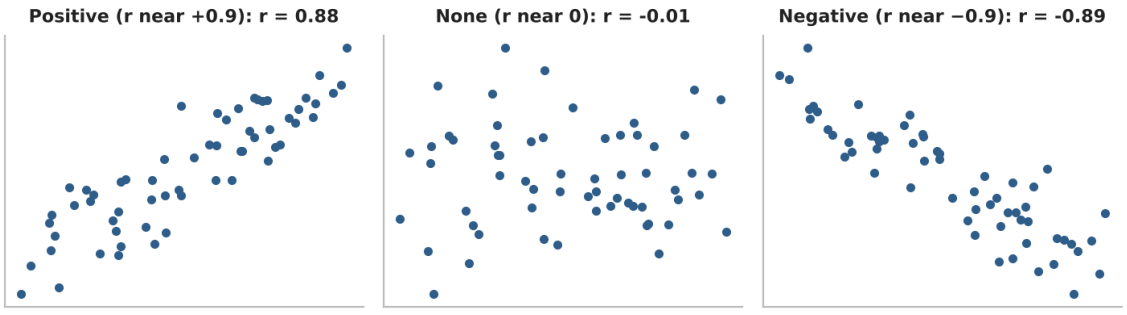


Figure A. What positive, no and negative correlation look like (simulated).

Value of r	Meaning
+1	Perfect positive linear relationship
Close to +1	Strong positive: when one rises, the other tends to rise
Around 0	No linear relationship
Close to −1	Strong negative: when one rises, the other tends to fall
−1	Perfect negative linear relationship

COMMON MISTAKE **r measures *linear* association only**

A perfect curve such as $y = x^2$ over a range symmetric about zero has $r = 0$ even though y is completely determined by x. Always look at the scatter plot as well.

3. The Vin Diesel example

The lecture uses three small data sets for the years 2010–2015: the number of films Vin Diesel made each year, US GDP growth and UK GDP growth.

Year	Vin Diesel films	US GDP growth (%)	UK GDP growth (%)
2010	0	2.53	1.54
2011	1	1.60	1.97
2012	3	2.22	1.18
2013	1	1.49	2.16
2014	4	2.43	2.85
2015	4	2.43	2.33

Pair	r	p-value (two-sided)	95% CI for ρ
Films vs US GDP growth	0.43	0.40	[−0.59, 0.92]
Films vs UK GDP growth	0.46	0.36	[−0.56, 0.93]
UK GDP growth vs US GDP growth	0.01	0.99	[−0.81, 0.81]

How to read the table: the p-value tests H_0: $\rho = 0$ (Topic 13 of the inferential course): a value above 0.05 means the correlation is not significant. The last column is a 95% confidence interval for the true correlation ρ.

The lecture reads these numbers as follows. Films and US growth correlate at 0.43; films and UK growth at 0.46; and US and UK growth hardly at all (0.01). Taken at face value this says Vin Diesel's film output is more related to the UK economy than the UK and US economies are to each other, which makes no sense.

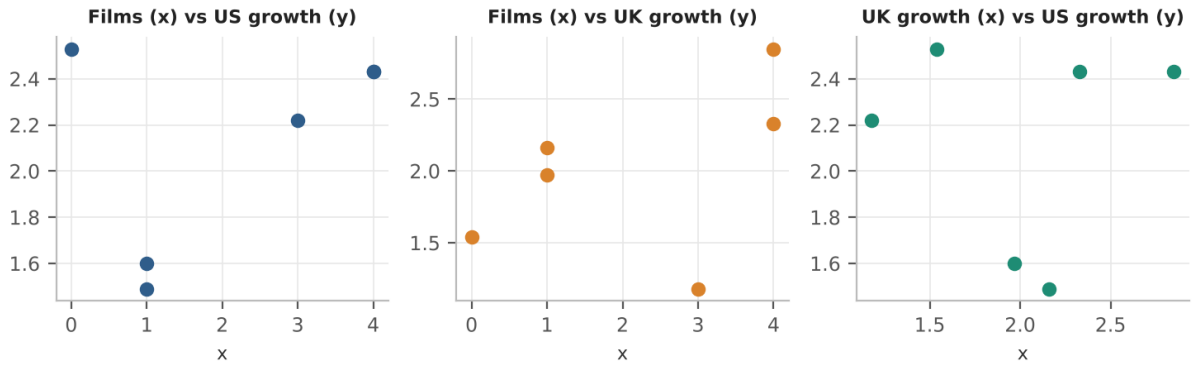


Figure 1. The three Vin Diesel scatter plots. With only six points, any pattern is fragile.

KEY IDEA **What the numbers really say**

- **Correlation does not imply causation.** Film output cannot plausibly drive GDP growth in two countries, and neither can the reverse.
- **The sample is tiny.** With n = 6, even r = 0.46 has a p-value of 0.36, far above 0.05. The 95% interval for ρ runs from about −0.6 to 0.9, so we cannot even say the correlation is positive.
- The class notes call the 0.46 correlation 'strong'; by common labels it is moderate, and by the test it is not significant.

In the lecture's own words: the result is a **coincidence**. It is easy to find a correlation; it is hard to find a causation.

The missing correlation between the two economies

The US–UK correlation is r = 0.008, about zero. But the sample covers only 2010–2015. A different period could give a different value, and trading partners usually have related economies. A low r in a short window is not proof of no relation. It is also a reminder that these six numbers are **a sample** and r is a **statistic**, as in the inferential topics.

4. Why a correlation can appear without causation

Reason	Meaning	Example
Coincidence	With few points or many variables, chance alone produces correlations	Vin Diesel films and GDP
Common cause (confounder)	A third variable drives both x and y	Hot weather raises both ice-cream sales and swimming accidents
Reverse causation	y actually affects x	Do hospitals cause illness? Sick people go to hospitals

Tyler Vigen's book and website *Spurious Correlations*, which the lecture workbook recommends, collects a large number of absurd but high correlations, such as unrelated yearly series that happen to drift together.

Demonstration with a confounder

Simulated summer data (200 days): temperature drives ice-cream sales and drowning incidents, but sales do **not** cause drownings. The raw correlation between sales and drownings is **0.64**. After removing the effect of temperature from both, the remaining correlation is **−0.00**, near zero. The link was entirely the common cause.

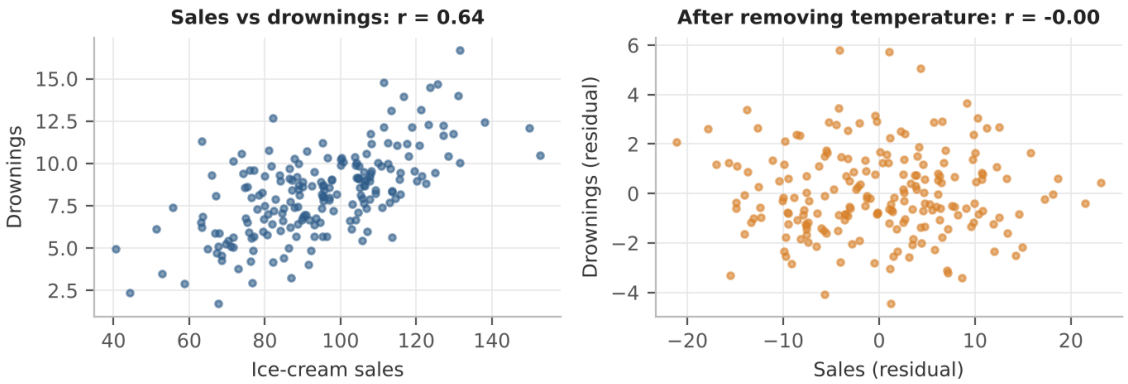


Figure 2. Simulated data. A strong raw correlation disappears once the common cause is controlled for.

5. What it takes to claim causation

- **Timing:** the cause comes before the effect.
- **Mechanism:** there is a believable reason x would change y.
- **No better explanation:** confounders are controlled or ruled out, ideally by a randomised experiment.
- **Replication:** the finding appears again in new data.

Regression can describe and predict. By itself it cannot prove causation; the logic of the study design has to supply that. The next topic shows how regression and correlation answer different questions.

6. In Python

```
import numpy as np
from scipy import stats

films = [0, 1, 3, 1, 4, 4]
uk    = [1.54, 1.97, 1.18, 2.16, 2.85, 2.33]

r, p = stats.pearsonr(films, uk)
print(round(r, 2), round(p, 2))           # 0.46  0.36

# 95% confidence interval for rho (Fisher z)
z = np.arctanh(r); se = 1 / np.sqrt(len(films) - 3)
print(np.tanh(z - 1.96 * se), np.tanh(z + 1.96 * se))
```

SUMMARY

- Regression explains or predicts a dependent variable y from one or more independent variables x.
- $r = s_{xy}/(s_x\, s_y)$ lies in $[-1, 1]$ and measures linear association.
- Correlation does not imply causation: coincidence, a common cause or reverse causation can create it.
- Very small samples (n = 6 here) give unstable correlations; test them and look at intervals.
- Causal claims need timing, mechanism, control of confounders and ideally an experiment.

Practice questions

Try each question before reading the answer underneath it.

Q1. Identify the dependent and independent variable: 'Does hours of sleep affect exam score?'

Answer. Dependent: exam score. Independent: hours of sleep.

Q2. Films and UK growth have $r = 0.46$ with $n = 6$. Is the correlation statistically significant at 5%?

Answer. No. $p = 0.36$, much greater than 0.05.

Q3. A study finds shoe size and reading score correlate strongly in primary school children. What is the likely common cause?

Answer. Age. Older children have bigger feet and read better.

Q4. Can $r = 0$ be given for variables that are perfectly related? Explain.

Answer. Yes. If the relationship is curved (such as $y = x^2$ over a symmetric range), r measures only linear association and can be 0.

Q5. Why is the US–UK correlation of 0.01 not proof that the two economies are unrelated?

Answer. It uses only six years (2010–2015), so it is a noisy estimate; another period or a different measure could give a different result.